

- 5 Two subspecies of reindeer, *Rangifer tarandus*, live in North America. Members of the different subspecies belong to the same species but have some morphological differences and are found in different geographical locations.

Fig. 5.1 shows a reindeer.



Fig. 5.1

Table 5.1 compares the features of the two North American reindeer subspecies.

Table 5.1

feature	woodland subspecies, <i>R. tarandus caribou</i>	barren ground subspecies, <i>R. tarandus groenlandicus</i>
habitat	southern woodland (warmer)	northern tundra (colder)
type of food	tree leaves, grass	lichens, moss
summer and winter feeding grounds overlap	yes	no
carry out long migrations	no	yes
body size	large	small
colour of fur	dark	light

- (a) During the last ice age an ice sheet separated southern and northern populations of *R. tarandus* in North America.

Explain how this ice sheet affected the evolution of *R. tarandus* to result in the two different subspecies.

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- (b) Assess the relative importance of natural selection and genetic drift in producing:

- (i) the different colours of fur of the two subspecies of reindeer

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- (ii) the different body sizes of the two subspecies of reindeer.

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- (c) Hybridisation has occurred between individuals of the two subspecies which now live in the area previously covered by the ice sheet.

Comment on how the hybrid populations compare to the pure subspecies in terms of genetic variation and potential to adapt to climate change.

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- (d) Outline how practical techniques could be used to test the hypothesis that migratory behaviour in reindeer has a genetic basis.

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